

Application of Information & Communication Technologies

Lecture-8

Recap of Lecture 7

- ◆ Storage System Characteristics
- ◆ Primary Storage
- ◆ Secondary Storage
 - HDD & SDD

Overview of Lecture 8

- ◆ Secondary Storage
 - Partitioning and File Systems
 - Optical Discs and Drives
- ◆ Network Cloud Storage
- ◆ Other types of Storage Systems

What is Partitioning?

- ◆ A hard disk is one physical device (example: 1 TB).
- ◆ Partitioning = dividing it logically into smaller sections.
- ◆ Each section acts like a separate drive (C:, D:, E:).
- ◆ It is not a physical cut, only a logical division of physical capacity.

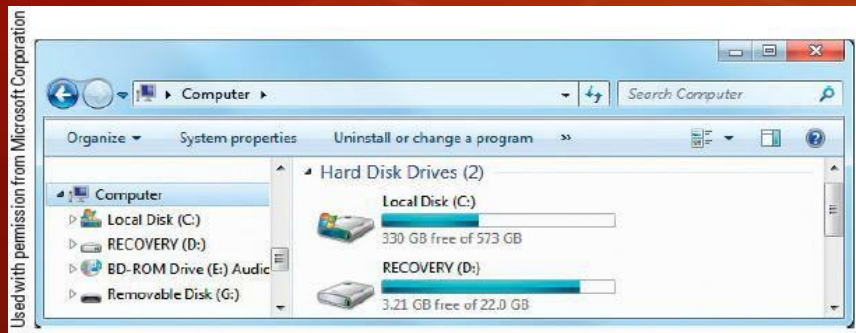


FIGURE 3-10
Hard drive partitions. New personal computers today often come with the primary hard drive divided into two partitions.

Why Partitioning?

- ◆ **Organize Data** → separate OS files, personal files, and backup.
- ◆ **Multiple Operating Systems** → Windows on C:, Linux on D:.
- ◆ **Backup & Recovery** → store recovery files in a safe partition.
- ◆ **Efficiency** → smaller sections are easier to manage.

File Systems

- ◆ **A file system** = set of rules for storing and organizing files on a drive.
- ◆ Tells computer **where data is saved** inside a partition.
- ◆ Helps in naming, finding, and managing files.

Example of File Systems

- ◆ **FAT32** = old system, limited (max 4GB). Used in USB, SD cards.
- ◆ **NTFS** = Modern, used in Windows, support very large file, security features(permission, encryption).
- ◆ **EXT4** = common in linux systems , fast and reliables.

File System Features

◆ Decides:

- Maximum file size
- Maximum partition size
- **Controls** how files are stored, retrieved (clusters/blocks)
- Security options (permissions, encryption)
- Different OS use different file systems.

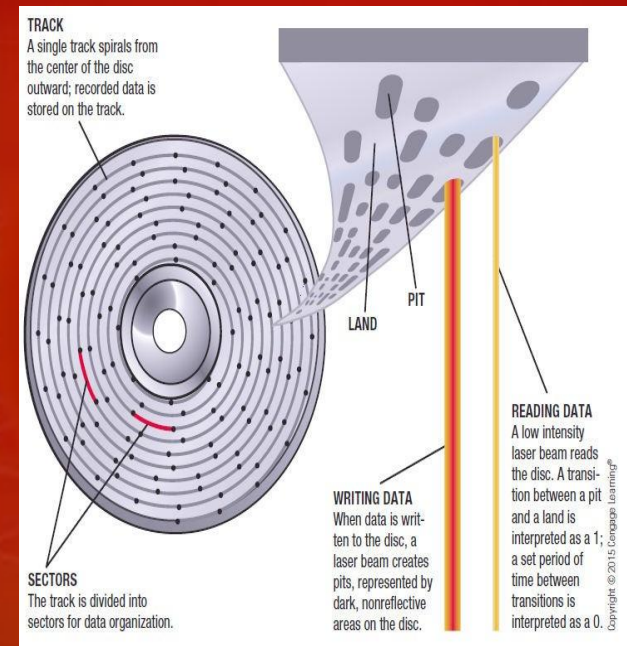
Optical Discs & Drives

- ◆ **Optical Discs:**
 - A type of storage read from and written to using laser beams
 - Examples: CD, DVD, Blu-ray (BD)
- ◆ Divided into sectors like magnetic discs but use a single spiral track (groove)
- ◆ Used for backup purposes and for storing and transporting music, photos, video, etc.

Types of Optical Discs

Representing Data on an Optical Disc

- Read-only optical disc
 - Surface of disc is molded or stamped
- Recordable or rewritable disc
 - Optical drive is used and the reflectivity of the disc is changed using a laser to represent the data
- Pits and lands are used to represent '1's and '0's
- The transition between a pit and a land represents a '1'; no transition represents a '0'



Capacity Comparison

TYPE OF DISC	CAPACITIY	USED FOR
CD	700 MB	Audio music delivery; custom CDs containing music, photos, etc.
DVD DVD-DL	4.7 GB 8.5 GB	Movie and software delivery; custom DVDs containing videos, music, photos, etc.
BD BD-DL BDXL	25 GB 50 GB 100 GB (rewritable) or 128 GB (recordable)	Primarily movie delivery

Advantages & Limitations

- ◆ Cheap, portable
- ◆ But: lower capacity, slower than SSD

Network and Cloud Storage Systems

- ◆ What is Network Storage?
 - Remote storage
 - Using a storage device not directly connected to the computer being used
 - Accessed through the Internet or through a network
 - Network storage
 - Remote storage via local network

Network and Cloud Storage Systems

- ◆ What is Cloud Storage?
 - Data stored on Internet servers and access via internet
 - Example: Google Drive, Dropbox, OneDrive

FIGURE 3-20

Cloud storage. This site provides 7 GB of free storage.



Features, Advantages & Issue of Cloud Storage

◆ Features:

- Access anywhere
- Sharing & collaboration
- Automatic backup

◆ Advantages

- Easy access ,
- Low cost for small users ,
- No hardware needed

◆ Issues with Cloud Storage

- Needs Internet,
- Security concerns & Limited free storage

Other Storage Systems

- ◆ Tape Storage:
 - Sequential access
 - Used for backup in data centers



Other Storage Systems

◆ Smart Card

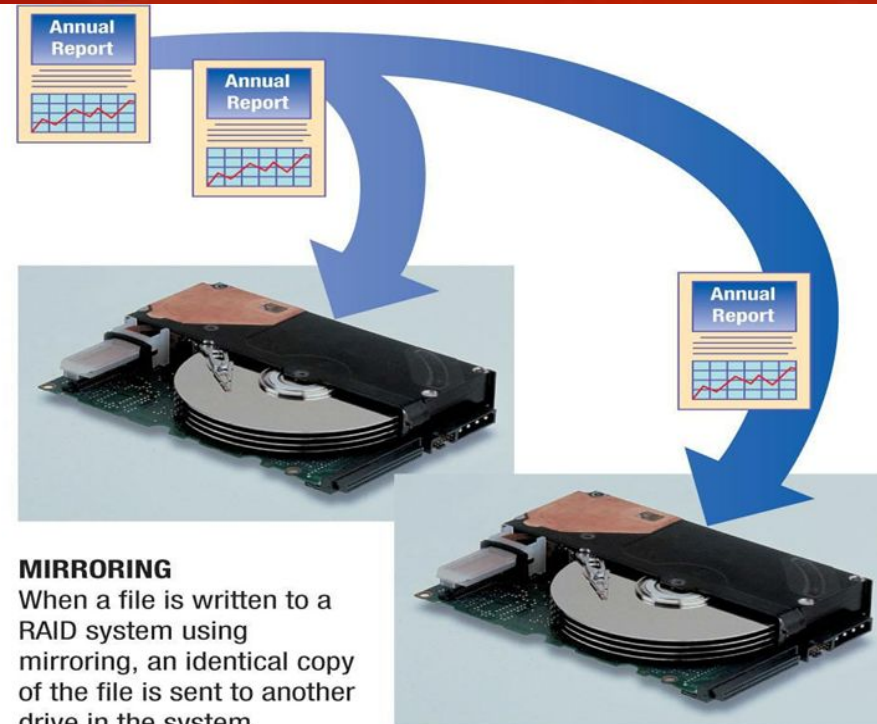
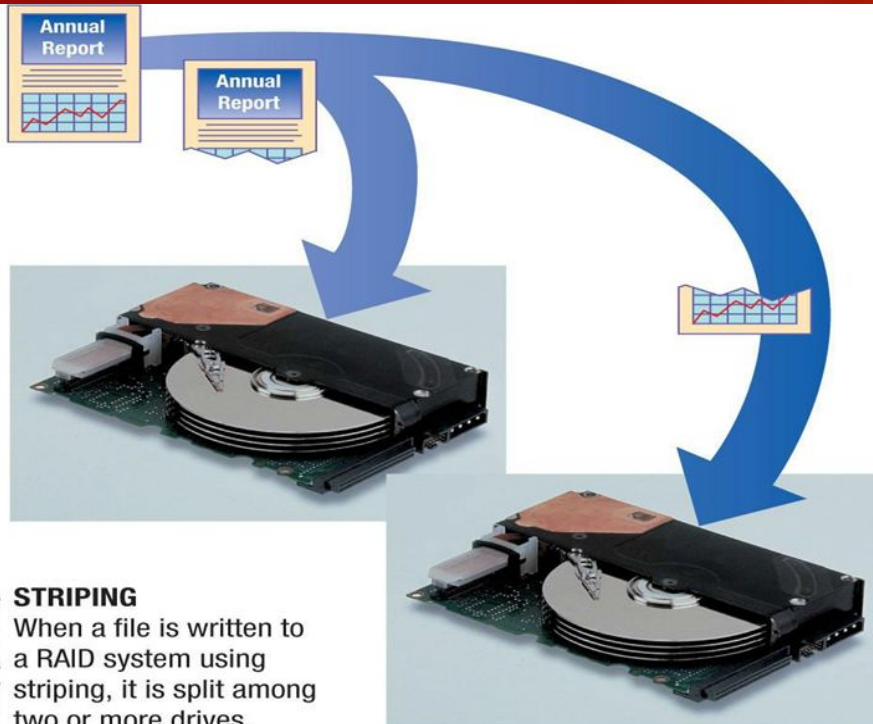
- Credit card-sized device with chip (processor + memory).
- Stores small data (like prepaid balance, ID info).
- Used in ATMs, door locks, SIM cards, transport cards.
- Some store biometric data (fingerprint, etc.).



FIGURE 3-21
Common smart card applications.

Other Storage Systems

- ◆ RAID Systems
 - Redundant Array of Independent Disks
 - Combines multiple drives for speed or safety



Other Storage Systems

- ◆ USB Flash Drive
- ◆ Also called: flash drive, thumb drive, jump drive.
- ◆ Small, portable, plug into USB port.
- ◆ Stores data (from MBs to GBs).
- ◆ Can include security features (passwords, fingerprint).



Courtesy Kingston Technology Company, Inc.

CONVENTIONAL DRIVE



Courtesy CustomUSB.com

CUSTOM CONVENTIONAL DRIVE



Courtesy CustomUSB.com

CUSTOM LANYARD DRIVE



Courtesy CustomUSB.com

CUSTOM WALLET DRIVE

Other Storage Systems

◆ Future Storage Trends

- **DNA Storage**  = “data inside biology.”
 - Uses DNA molecules to store digital data.
 - Extremely high capacity
 - 1 gram of DNA could store hundreds of terabytes.
- **Holographic Storage**  = “data stored with laser light in 3D.”
 - Stores data in 3D patterns of light inside crystals or polymers.
 - Can store more data in less space than CDs/DVDs.
 - Much faster read/write.
- **3D NAND** = “modern SSD technology”
 - Advanced flash memory (used in SSDs).
 - Stores data in vertical layers (not flat), saving more space.
 - Modern SSDs with TBs of capacity use this.

Summary

- ◆ Secondary Storage
 - Partitioning and File Systems
 - Optical Discs and Drives
- ◆ Network Cloud Storage
- ◆ Other types of Storage Systems

Suggested Reading

- ◆ Ch-03, The System Unit: Processing and Memory ,
“Understanding Computers: Today and Tomorrow,
Comprehensive”, 15th Edition by Deborah Morley
& Charles S. Parker
- ◆ Ch-06, Discovering Computers Fundamentals-
Your Interactive Guide -- Gary B Shelly; Misty E
Vermaat; Jeffrey J Q